2025.12.1 HW9 习题五.

B9. $P(0.95<X<1.05)=0.0975$

(1) 故 $Y\sim B(100,0.0975)$

i) 精确分布: $P(Y>2)=1-(0.9025)^{98}[(0.9025)^2+9.75\times 0.9025+(0.0975^2\times 4950)]=0.9976$.

ii) 泊松分布: $\lambda=100\times 0.0975=9.75$

$$P(Y>2)=1-P(Y\le 2)=1-\left(e^{-9.75}+9.75e^{-9.75}+\frac{e^{-9.75}\cdot 9.75^2}{2}\right)=0.9966.$$

iii) 中心极限定理: $\dfrac{Y-100\times 0.0975}{\sqrt{100\times 0.0975\times 0.9025}}=\dfrac{Y-9.75}{2.966}\sim N(0,1)$

$\therefore P(Y>2)=1-\Phi(-2.61)=\Phi(2.61)=0.9955$.

(2) $P(\frac{1}{2}<X<\frac{3}{2})=0.75$, 故 $Z\sim B(n,0.75)$.

$$\frac{Z-0.75n}{\sqrt{n\cdot 0.75\cdot 0.25}}=\frac{Z-0.75n}{\sqrt{0.1875n}}\sim N(0,1).$$

故 $P(Z \geqslant 80) = 1 - \Phi\left(\frac{80-0.75n}{\sqrt{0.1875n}}\right) \geqslant 0.95$

$\Rightarrow n \geqslant 117$. $\therefore n$ 最小为 117.

B10. $P(X_1=0)=0.2$, $P(X_1=1)=0.3$, $P(X_1=2)=0.5$

$\therefore E(X_1)=1.3$, $Var(X_1)=0.61$

由中心极限定理,

$$P\left(\sum_{i=1}^{800} X_i > 1000\right) = \Phi\left(-\frac{1000-1040}{\sqrt{0.61\times 800}}\right) = 0.9649.$$

B11. (1) $n_A \sim B(100, 0.3)$.

记得0分的人数为 n_A.

$$\frac{n_A - 100\times 0.3}{\sqrt{100\times 0.3\times 0.7}} \sim N(0,1).$$

$\therefore P(A \leqslant 35) = \Phi\left(\frac{5}{\sqrt{21}}\right) = 0.8621$

(2) $E(X_1) = 0\times 0.2 + 1\times 0.16 + 2\times 0.128 + 4\times 0.512 = 2.464$

$E(X_1^2) = 8.864$, $Var(X_1) = 2.793$.

$$\therefore \frac{\sum_{i=1}^{100} X_i - 2.464\times 100}{\sqrt{2.793\times 100}} \sim N(0,1).$$

故 $P\left(\sum_{i=1}^{100} X_i > 220\right) = \Phi\left(\frac{26.4}{\sqrt{279.3}}\right) = 0.9429$

习题六 A5.

(1) $E(X) = 1\times 0.3 = 0.3$. $\therefore E(\overline{X}) = 0.3$

$Var(X) = 1\times 0.3\times 0.7 = 0.21$ $\therefore Var(\overline{X}) = \frac{0.21}{7} = 0.03$

(2) $E(S^2) = 0.21$

(3) $P(\max(X_1 \dots X_7) < 1) = 0.7^7$

B4. $\overline{X} \sim N(a, \frac{2.5^2}{n})$

故 $\frac{\overline{X}-a}{\frac{2.5}{\sqrt{n}}} \sim N(0,1)$

$P(|\overline{X}-a| \leq 0.5) = 1 - 2\Phi(\frac{-\sqrt{n}}{5})$

(1) $\Phi(\frac{-\sqrt{n}}{5}) \leq 0.05$

$-\frac{\sqrt{n}}{5} \leq -1.645 \Rightarrow n \geq 68$

(2) $\Phi(-\frac{\sqrt{n}}{5}) \leq 0.025$

$-\frac{\sqrt{n}}{5} \leq -1.96 \Rightarrow n \geq 97$.

B6. $E(X)=\frac{\theta}{2}$，$E(X^2)=\frac{\theta^2}{3}$，$Var(X)=\frac{\theta^2}{12}$

$\therefore E(\bar{X})=\frac{\theta}{2}$ $Var(\bar{X})=\frac{\theta^2}{60}$. $E(\bar{X}^2)=\frac{\theta^2}{60}+\frac{\theta^2}{4}=\frac{4\theta^2}{15}$.

$E(S^2)=\frac{\theta^2}{12}$.

B7. (1) $E(X)=\frac{1}{\lambda}$，$Var(X)=\frac{1}{\lambda^2}$.

$\therefore E(\bar{X})=\frac{1}{\lambda}$，$Var(\bar{X})=\frac{1}{10\lambda^2}$.

(2) $P(X_1>k, X_2>k \cdots X_{10}>k)=\prod_{i=1}^{10} P(X_i>k)=(e^{-\lambda k})^{10}$
$=e^{-10\lambda k}$

故 $F_{X_{(1)}}(k)=1-e^{-10\lambda k}$.

故 $X_{(1)}\sim Exp(10\lambda)$.

$\therefore E(X_{(1)})=\frac{1}{10\lambda}$，$Var(X_{(1)})=\frac{1}{100\lambda^2}$.